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RESEARCH PAPER

Large Marks-decahedral Pd nanoparticles synthesized by a modified hydrothermal method using a homogeneous reactor

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Abstract Fivefold symmetry appears only in small particles and quasicrystals because internal stress in the particles increases with the particle size. However, a typical Marks decahedron with five re-entrant grooves located at the ends of the twin boundaries can further reduce the strain energy. During hydrothermal synthesis, it is difficult to stir the reaction solution contained in a digestion high-pressure tank because of the relatively small size and high-temperature and high-pressure sealed environment. In this work, we optimized a hydrothermal reaction system by replacing the conventional drying oven with a homogeneous reactor to shift the original static reaction solution into a full mixing state. Large Marks-decahedral Pd nanoparticles (~90 nm) have been successfully synthesized in the optimized hydrothermal synthesis system. Additionally, in the products, round Marks-decahedral Pd particles were also found for the first time. While it remains a challenge to understand the growth mechanism of the fivefold twinned structure, we proposed a plausible growth-mediated mechanism for Marks-decahedral Pd nanoparticles based on observations of the synthesis process.

Keywords Marks decahedron · Hydrothermal method · Growth mechanism · Nanoparticle growth model

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Introduction

Pd nanoparticles have attracted much attention because of their excellent performance in catalysis, including in the oxygen reduction reaction (ORR) (Shao et al. 2011), methanol oxidation reaction (MOR) (Liu et al. 2011), formic acid oxidation (FAO) (Jin et al. 2012), and carbon-carbon coupling reaction (Li and El-Sayed 2001). Like other noble metals, such as Ag and Au, shape control has emerged as one of the most effective strategies to optimize their catalytic performance (Zhang et al. 2013). The decahedral shapes including the regular decahedron (Huang et al. 2014), star-shaped decahedron (Lim et al. 2009), and Marks decahedron (Ji et al. 2014, 2015) shown in Fig. 1(a)–(c) have been prepared successfully by many groups. The round Marks decahedron shown in Fig. 1(d) was first synthesized in the present work.

Of the different decahedron shapes shown in Fig. 1, when their size is between 1 and 5 nm, the regular shape is rarely observed, and the most common shape is the Marks decahedron when the experiments are conducted in very

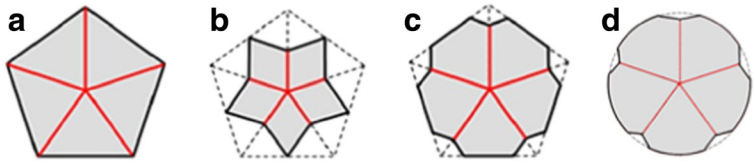


Fig. 1 Schematic drawings of **a** a regular decahedron, **b** a star-shaped decahedron, **c** a Marks decahedron, and **d** a round Marks decahedron

clean growth conditions (Yacamán et al. 2001). The stability of small decahedrons has been characterized by generalizing the surface structure of the nanoparticles using an improved Wulff construction (Marks 1983, 1984), i.e., a variant of the Wulff construction appropriate for twinned nanoparticles. It is generally accepted that a fivefold symmetry structure is allowed only in small particles and quasicrystals because the internal stresses caused by twinning increase as structures become larger (Shechtman et al. 1984). In a synthesis, depending on the particle size, the decahedron can also be favored because surface energy from the extra strain energy caused by twinning can be compensated for by maximizing the surface coverage with {111} facets to achieve the lowest surface energy (Elechiguerra et al. 2006; Lofton and Sigmund 2005). Later, the existence of decahedral nanoparticles up to intermediate sizes was rationalized (Ringe et al. 2013) by coupling the modified Wulff shapes with a disclination strain model for decahedral nanoparticles (De Wit 1972) and a 3D variant for icosahedral nanoparticles. Particularly, a typical Marks decahedron can avoid the strain energy concentrating in the tips of the pentagon by the production of five re-entrant grooves localized at the ends of the twin boundaries. This strain relief mechanism makes Marks decahedrons more stable than regular decahedrons and enables us to prepare large Marks decahedrons. In this paper, we will report our recent work on size-controlled Marks-decahedral Pd nanoparticles ranging from ~20 to ~90 nm.

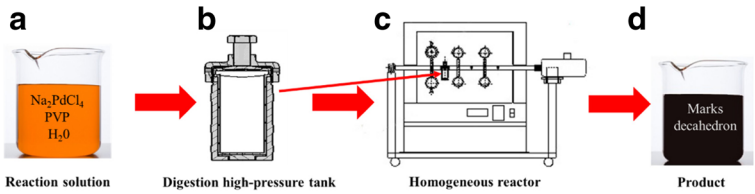
Experimental details

Palladium chloride ($PdCl_2$, AR), sodium tetrachloropalladate (Na_2PdCl_4, AR), and poly(vinyl pyrrolidone) (PVP = 55,000) were purchased from Aladdin Industrial Corporation (Shanghai, China). Deionized (DI) water was lab-made. All reagents were used without further purification.

In a standard synthesis, Marks-decahedral Pd nanoparticles were synthesized using a hydrothermal growth method, where Na_2PdCl_4, PVP, and DI water served as the metal precursor, stabilizer, and solvent, respectively, with PVP also used as a weak reductant because of its hydroxy (OH) end groups (Washio et al. 2006). A typical synthetic process is shown in Scheme 1. For the preparation of large Marks-decahedral Pd nanoparticles, first, 300.0 mg of PVP (=55,000) and 15.0 mL of high-purity DI water were introduced into a 20.0 mL vial under magnetic stirring for 10 min at room temperature. Subsequently, 33.0 mg (0.1 mmol) of Na_2PdCl_4 was added into the mixture with stirring for another 5 min. The orange-red solution was transferred to a 20.0 mL Teflon-lined stainless-steel high-pressure tank (Scheme 1(b)). After the temperature in the homogeneous reactor (Scheme 1(c)) reached 160 °C, the high-pressure tank was placed into the homogeneous reactor and maintained at 160 °C for 16 h. After the reaction is completed, the high-pressure tank was cooled naturally at room temperature. Finally, the obtained taupe Pd nanoparticle solution (Scheme 1(d)) was centrifuged (12,000 rpm, 30 min) and washed three times with DI water to remove excess PVP. The collected final products were dispersed and preserved in ethanol.

For the synthesis of round Marks-decahedral Pd nanoparticles, the experimental condition was the same as in the synthesis of the large Marks decahedrons except for decreasing the heating time to 12 h.

Transmission electron microscopy (TEM) images of the Marks-decahedral Pd nanoparticles were obtained using a Tecnai G2F20 microscope operating at an



Scheme 1 Schematic illustration of the synthetic pathway leading to the formation of Marks-decahedral Pd nanoparticles

accelerating voltage of 200 kV. High-resolution TEM (HRTEM) images were captured using a Titan G2 60-300 with an image corrector (point resolution = 80 pm) operating at an accelerating voltage of 300 kV. The samples for TEM and HRTEM were prepared by dropping the ethanol suspensions of the products on copper grids coated with amorphous carbon film.

Results and discussion

In general, Marks decahedrons can be considered to be the junction of five face-centered-cubic (FCC) truncated tetrahedral subunits bonded together by twin boundaries and closed by {111} crystallographic faces. Notably, the theoretical predicted angle between two planes of a truncated tetrahedral subunit is 70.53°, and the two planes of each re-entrant groove are parallel to their second nearby twin boundaries because they belong to the same facets as the boundaries, i.e., the {111} facets. Thus, the angle of each re-entrant groove is 141.06°, and the truncated tetrahedrons do not assemble to a completely space-filling structure, leaving an angle of 7.35° unclosed. The pseudo-fivefold symmetrical model is shown in Fig. 2(a). An intact TEM image of an as-prepared Pd nanoparticle is shown in Fig. 2(b). The pentagonal profile and the re-entrant grooves localized at the ends of the twin boundaries can be clearly observed, thus confirming the structure as a Marks decahedron. According to the simulation, the Marks decahedron is elastically strained by ~2% to form this space-filling structure (Marks and Smith 1981; Smith and Marks 1981). The stability of Marks-decahedral nanoparticles can be characterized by the inhomogeneous strain model (Ji et al. 2015; Johnson et al. 2008). The model agrees that a wedge disclination exists in the crystal. The single disclination coinciding with the fivefold axis removes the solid-angle deficiency, producing an inhomogeneous strain field and reducing the strain-energy density significantly. In Fig. 2(b), the measured individual angles of the re-entrant grooves are 137.62°, 133.12°, 148.72°, 142.78°, and 144.49°, which are inconsistent with the fivefold symmetrical model. This discrepancy reveals the presence of lattice distortion in the growth of Marks-decahedral Pd nanoparticles. The lattice deformation in the center of the nanoparticles can also be observed in Fig. 2(b). All of these observations agree fairly well with the inhomogeneous strain model.

We further characterized the Marks decahedrons by HRTEM. An HRTEM image recorded along the [011]

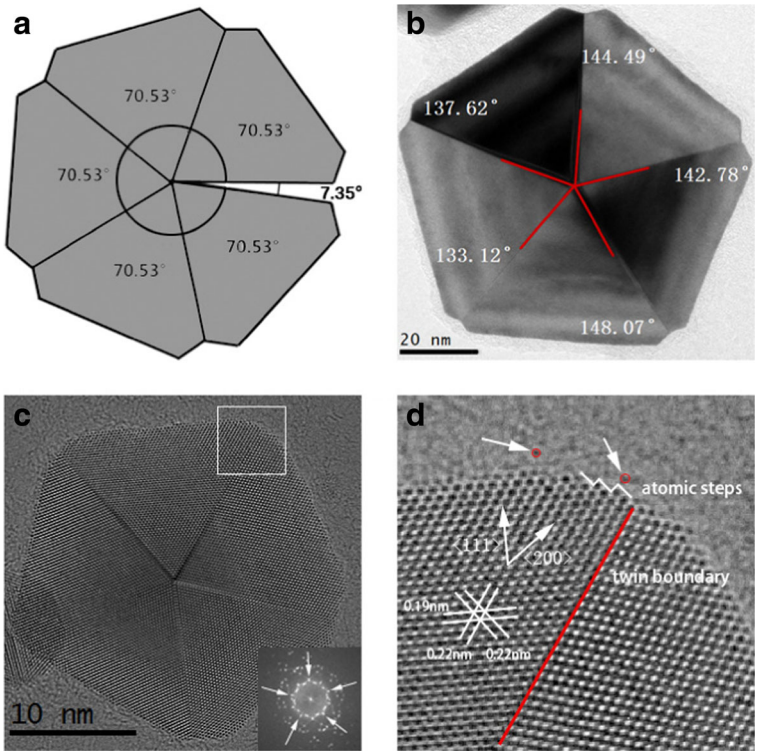


Fig. 2 **a** Geometric model of a Marks-decahedral nanoparticle with an angular deficiency of 7.35° (modified with permission from Ref. 9). **b** A typical TEM image of a Marks-decahedral Pd nanoparticle with a size of 82.83 nm. The angle of each re-entrant groove is noted around the particle, and the fivefold twinned structures are noted with red lines. **c** HRTEM image of an individual Marks-decahedral Pd nanoparticle and the corresponding FFT pattern (inset). **d** HRTEM image taken from the re-entrant groove noted with a white square in **c**, which shows the essential crystal crystallography of the Marks-decahedral Pd nanoparticle, including the twin boundaries, lattice spacing, crystal orientation, and atomic steps in the re-entrant groove (Color figure online)

zone axis is shown in Fig. 2(c). The fivefold symmetry structure and rough edges can be observed clearly. The inset is a fast Fourier transform (FFT) pattern of the high-resolution image, and the spots indicated by the arrows can be indexed as the {200} reflections from the five single-crystalline domains. The partial enlarged view of the re-entrant groove shown in Fig. 2(d), marked with a square in Fig. 2(c), shows the essential crystal crystallography of the Marks decahedron. The fringes with lattice spacings of 0.22 and 0.19 nm can be indexed as the {111} and {200} crystallographic planes of bulk FCC Pd, respectively. Some free Pd atoms marked with red circles and corresponding active sites for atomic deposition were also clearly observed. The atomic steps in the re-entrant grooves are marked with a white polyline in Fig. 2(d).

As-synthesized nanoparticles were deposited on a carbon layer to make the nanoparticles more stable under an electron beam and obtain selected-area electron diffraction (SAED) patterns in different areas. Figure 3(a), (d), and (g) shows the TEM images of different views of a Marks-decahedral Pd nanoparticle, Fig. 3(b), (e), and (h) shows the corresponding atomic models, and the corresponding SAED patterns are shown in Fig. 3(c), (f), and (i), respectively. The electron diffraction pattern in Fig. 3(c) was obtained by electron beam irradiation in the center of the crystal, indicating a fivefold symmetry structure with the fivefold axis equal to the incident beam direction. Six diffraction spots, noted with white circles, indicate a typical set of FCC single-crystal diffraction patterns recorded along the [011] zone axis. All diffraction reflections in Fig. 3(c) can be indexed by the overlapping of twin-related five single-crystal diffraction patterns, as indicated by the five parallelograms marked with different colors. Each set of diffraction patterns can be obtained by the rotation of another set of diffraction patterns by $n \times 72°$ ($n = 1, 2, 3, 4$). Figure 3(f) is an SAED pattern recorded along the [110] zone axis with a shape of a flat hexagon. The intersection angle between the {111} and {200} crystallographic planes is 54.74°, which is in good agreement with the FCC single-crystal diffraction pattern of the [110] zone axis. Likewise, in Fig. 3(i), the 60° angle between {220} planes and the regular hexagonal shape denote the SAED pattern of an FCC single-crystal recorded along the [111] zone axis.

In our previous work (Ji et al. 2014), the effects of temperature and PVP on constructing the shapes of Marks decahedrons were investigated in detail. While

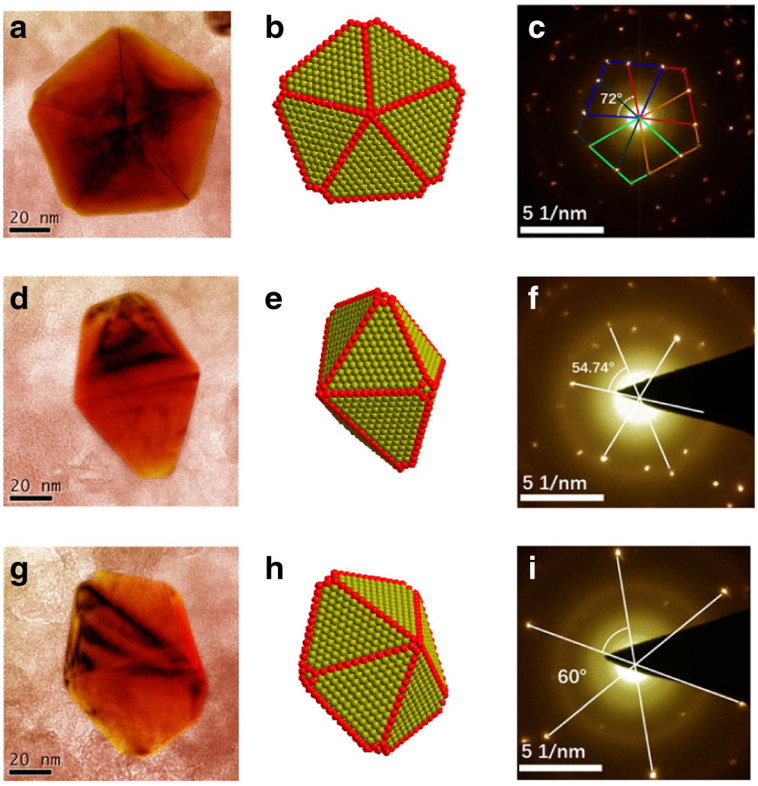


Fig. 3 **a**, **d**, and **g** are TEM images of Marks decahedrons obtained at different angles, and **b**, **e**, and **h** are the corresponding atomic models; **c**, **f**, and **i** are the selected-area electron diffraction patterns related to **a**, **d**, and **g**, respectively

the yield of Marks decahedrons is not proportional to temperature, an appropriate temperature is instrumental to achieving a desirable yield rate. Increasing the molecular weight of PVP plays a more important role on the size of Marks decahedrons than the mass ratio of the metal precursor to PVP. Herein, to control the size of Marks-decahedral Pd nanoparticles, we focus on the influence factors of precursor type and heating method.

To determine the role of the Pd precursor in controlling the size of Marks decahedrons, we conducted a series of synthesis experiments replacing the 20.0 mg (0.1 mmol) of $PdCl_2$ with 33.0 mg (0.1 mmol) of Na_2PdCl_4. Figure 4(a) and (b) shows the TEM images of samples synthesized in a conventional autoclave with different precursors. All the experimental conditions are the same except for the precursor. With $PdCl_2$ used as the precursor, the Marks decahedrons have an average edge length of ~30 nm, and the products are found to contain ~35% Marks decahedrons. With Na_2PdCl_4, the products contain ~50% Marks decahedrons and regular decahedrons, and the size of the Marks decahedrons increased to ~70 nm. The nucleation and growth of nanoparticles depend critically on the reduction rate of the metal cation or, equally, the rate at which metal atoms are generated in the solution. This process is governed by the rate at which electrons are transferred

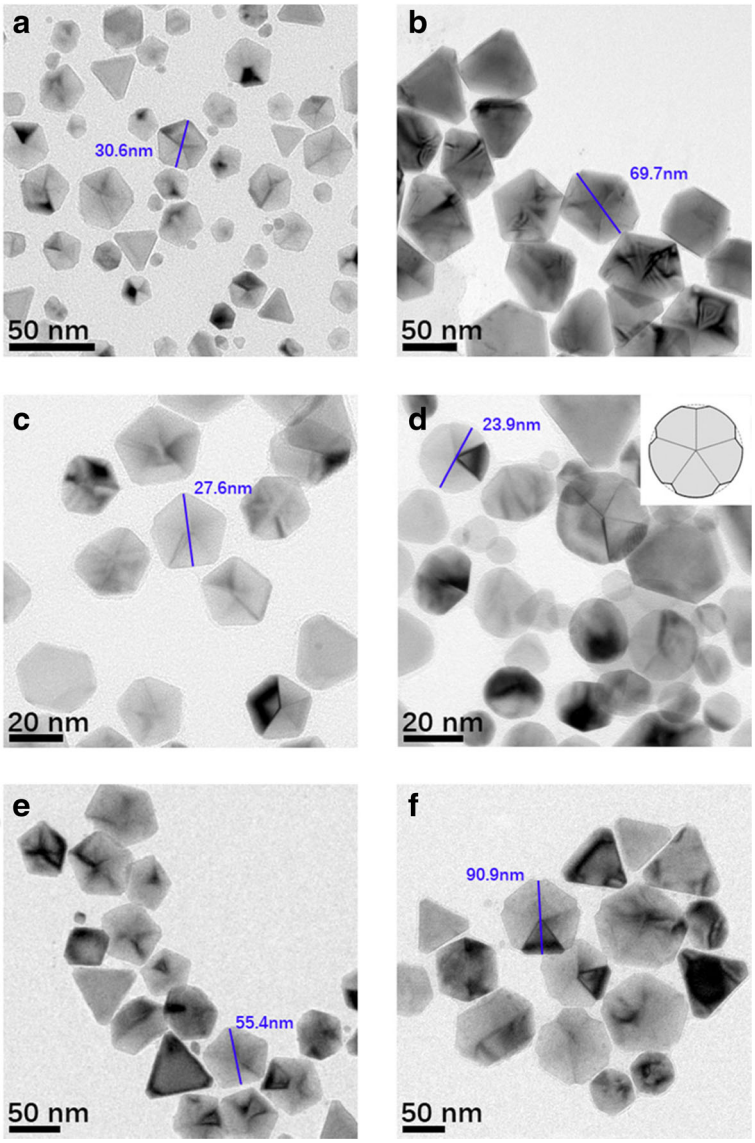


Fig. 4 TEM images of Marks-decahedral Pd nanoparticles prepared under different conditions. Heating 15.0 mL of an aqueous solution containing 20.0 mg of $PdCl_2$ (**a**) or 33.0 mg of Na_2PdCl_4 (**b**) with 300.0 mg of PVP (=55,000) for 16 h in a conventional drying oven. Heating 15.0 mL of an aqueous solution containing 33.0 mg of Na_2PdCl_4 and 300.0 mg of PVP (=55,000) for 12 h (**c**, **d**), 14 h (**e**), or 16 h (**f**) in a homogeneous reactor. As-synthesized round Marks decahedrons and their corresponding geometric model (inset) are shown in **d**

from the reductant to the metal precursor. Based on Marcus theory, introducing the fundamental physics of electron transfer between molecular species, the familiar relation between the reaction rate constant k and the difference in Gibbs free energy (ΔG) of the reaction is known (Marcus 1993):

$$k = A \cdot e^{-\Delta G/k_B T} \quad (1)$$

where A depends on the characteristic value of the electron transfer process, k_B is the Boltzmann constant, and T is the absolute temperature. According to this formula, the kinetics of an electron transfer process and thus the rate of reduction are intimately linked to the thermodynamic stability of the reactants. It is easy to understand that metal ions with more positive reduction potentials tend to be reduced at a faster rate than those with lower potentials (Gilroy et al. 2016). Thus, the lower reduction potential of $PdCl_4^{2+}$/Pd (0.62 V vs RHE) compared to Pd^{2+}/Pd (0.95 V vs RHE) explains this increased-size phenomenon of the as-synthesized Marks-decahedral Pd nanoparticles. The standard reduction potential of Na_2PdCl_4 is one third that of $PdCl_2$, and the reduction rate would follow the same trend: $PdCl_4^{2+} < Pd^{2+}$. In the same experimental conditions, the generation rate of Pd atoms is slowed, with consequentially small quantities of Marks-decahedral Pd seeds, when Na_2PdCl_4 is adopted as the Pd precursor. At the same concentration of precursor, the presence of fewer seeds means larger sizes of the final product.

It is very difficult to stir reaction solutions contained in digestion high-pressure tanks due to their relatively small size and high-temperature, high-pressure sealed environments. Thus, we attempted to optimize the hydrothermal reaction system by replacing the conventional drying oven with a homogeneous reactor to heat the reaction solution. The homogeneous reactor is assembled from an incubator box with a rotating shaft and a motor. The hydrothermal synthesis reaction vessel is fixed on the rotating shaft through the collar, as shown in Scheme 1(c), which shifts the original static reaction solution into a full mixing state and effectively promotes the reaction. Figure 4(b) and (f) shows the TEM images of Marks-decahedral Pd samples obtained by heating 15.0 mL of an aqueous solution containing 33.0 mg of Na_2PdCl_4 and 300.0 mg of PVP (=55,000) for 16 h in a conventional drying oven and the homogeneous reactor, respectively. The products heated in the conventional drying oven have an average length of ~70 nm. It is worth mentioning that the nanoparticles produced in the homogeneous reactor have a more even size of ~90 nm with five distinct re-entrant grooves. The size of the as-synthesized Marks decahedrons is three times larger than our previous synthesized products using a conventional drying oven (as in Fig. 4(a)). It is also worth noting that a change of heating pattern can effectively increase the size of Marks-decahedral Pd nanoparticles synthesized in hydrothermal solutions and create monodisperse nanoparticles. This observation can probably be attributed to the full stirring effect implemented by the homogeneous reactor. Sufficiently mixing the reaction solution increases the probability of particle collision and results in nuclei fused into larger and more uniform sizes. In this reaction solution, full mixing also increases the chance of contact between the atoms and the crystal surface, providing adequate Pd atoms for the growth of Pd nanoparticles, further enlarging the nanoparticle size.

The heating time is another pivotal influencing factor of the hydrothermal synthesis reaction. To understand its role, we examined different reaction times. Figure 4(c), (d), (e), and (f) shows the TEM images of Pd samples obtained by heating 15.0 mL aqueous solutions (33.0 mg Na_2PdCl_4, 300.0 mg PVP, 160 °C) for different times. After heating for 12 h, Fig. 4(c) and (d), Marks decahedrons have an average size of ~25 nm. An interesting and meaningful finding in Fig. 4(d) is that round Marks-decahedral Pd nanoparticles were produced under this reaction condition. To our best knowledge, this may be the first report synthesizing round Marks-decahedral Pd nanoparticles in a hydrothermal synthesis system. The facets corresponding to the twin boundaries on a decahedron can be affirmed as {211}, while the facets corresponding to the edge can be affirmed as {100} (Choi et al. 2015). It is reported that the surface energy of Pd (100) (1661 ergs/cm^2) is slightly higher than that of Pd (211) (1617 ergs/cm^2), indicating that (111) (1382 ergs/cm^2) is much lower than both (100) and (211) in terms of surface energy (Zhang et al. 2004). In addition, the rate of diffusion of an adatom across a surface is determined by the diffusion coefficient D, which can be expressed as (Xia et al. 2013):

$$D = D_0 \cdot e^{-E_{diff}/RT} \quad (2)$$

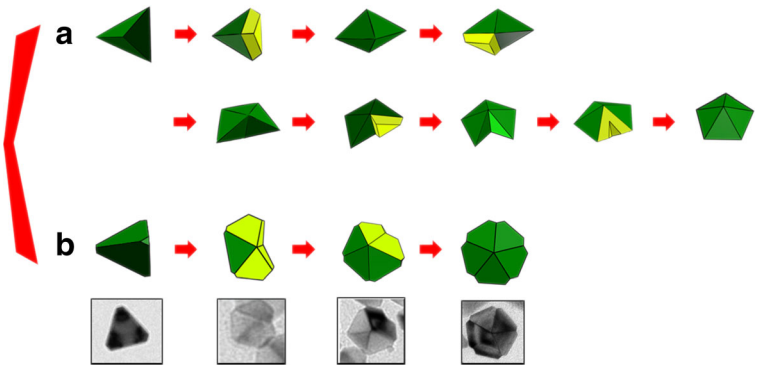
where D_0 is the pre-exponential factor, E_{diff} is the energy barrier to diffusion, R is the ideal gas constant, and T is the absolute temperature. Therefore, at 160 °C, the Pd atoms deposited at vertices, tending to diffuse along the edges and

then remain at these sites and form rounder edges to minimize both the internal strain and the total surface energy (Yacaman et al. 2001). When reacted for 14 h, the particle size increased to ~50 nm, as in Fig. 4(e). We further prolonged the heating time to 16 h, long enough for Pd atoms to diffuse over the entire surface of the Marks decahedron. The essence of thermodynamic control is to give a system an adequately long period so that all the atoms can reach their final destinations to attain a globally minimized Gibbs free energy of the entire system (Xia et al. 2015). Thus, heating the sample for a long time, thereby raising the temperature to increase the mobility of atoms on the surface, can effectively increase the crystal size. It is worth pointing out that Ostwald ripening also plays an important role in controlling the growth process.

The growth of nanoparticles is controlled by thermodynamics and kinetics, with thermodynamics granting possible stable structures (Qi 2016) and growth kinetics specifying the real formation route. Recent experimental and theoretical studies have shown that particle coalescence can also play an important role in nanoparticle growth (Zheng and Alivisatos 2009). The formation of a decahedron occurs through an edge-selective particle fusion mechanism, with five tetrahedrons gradually assembling into a decahedron (Gao et al. 2006). Recently, Masaharu Tsuji and co-workers proposed that decahedrons grow through one-by-one stepwise growth of tetrahedral units without the stepwise fusion of twin facets (Tsuji et al. 2010), as shown in Scheme 2(A). Furthermore, they thought that the nanoparticle growth occurs on only one facet until the formation of a truncated tetra-tetrahedron, and the fifth tetrahedron is grown from the two facing {111} planes and is connected in the intermediate position. In consideration of the fivefold symmetry of the Marks decahedron and the observation of a series of intermediate particles with imperfect truncated tetrahedral units, as shown in Scheme 2(B), the stepwise growth mechanism for Marks decahedrons is supported. In the TEM images, we found the simultaneous growth of two facets of a truncated single tetrahedron and a truncated tri-tetrahedron, which is different from the aforementioned one-by-one stepwise growth mechanism for regular decahedrons. Based on these observations, we propose a modified growth mechanism for the Marks-decahedral structure. The growth route is shown in Scheme 2(B). First, many truncated tetrahedrons are formed initially in the reaction solution. Their exposed {111} crystallographic planes provide enormous numbers of active sites for atomic deposition and epitaxial growth. Then, two adjacent {111} facets of a truncated single tetrahedron simultaneously grow epitaxially, forming a truncated tri-tetrahedron. Without the formation of a truncated tetra-tetrahedron, a Marks decahedron is formed directly by epitaxial growth of both concave {111} facets of a truncated tri-tetrahedron.

Conclusions

In summary, a modified hydrothermal method has been developed to control the formation of size-controlled Marks-decahedral Pd nanoparticles. A robust way of synthesizing large nanoparticles is heating 15.0 mL of an aqueous solution containing 33.0 mg of Na_2PdCl_4 and 300.0 mg of PVP simultaneously in a homogeneous reactor for 16 h. The types of metal precursors and the heating method are critical for the nucleation process. The slow reduction rate of Na_2PdCl_4, the mixing homogeneity of the reaction solution, and the long reaction time are necessary to grow large Pd nanoparticles. Additionally, round Marks-decahedral Pd particles were found for the first time during the synthesis process; they are thermodynamically stable because a nearly



Scheme 2 Growth mechanisms of five-twinned decahedral nanoparticles. **a** A regular decahedron is formed via one-by-one growth of tetrahedral units. **b** A modified stepwise growth mechanism for Marks-decahedral Pd nanoparticles. Yellow parts show the growing structures of the truncated tetrahedron (Color figure online)

spherical profile can achieve a lower surface energy. We also proposed a plausible growth mechanism for the formation of Marks-decahedral Pd nanoparticles.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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